GENERATIVE AI: PRACTICAL SESSION

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Environment

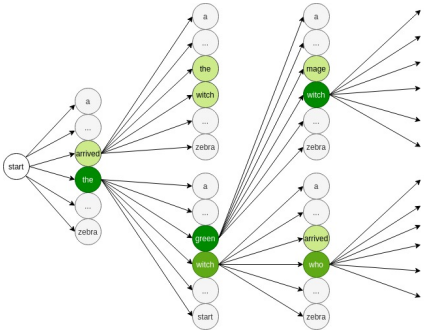
1. To play quickly with LLM: `https://textsynth.com/completion.html`
   - Distinguish LLM with/wo instruction
   - Understand temperature
2. Resolve real use case with real LLMs (will require an -free- account)
   - chatGPT : `https://chatgpt.com`
   - Claude : `https://claude.ai/new`
   - Google Gemini : `https://gemini.google.com/app`
   - Mistral : `https://chat.mistral.ai/chat`
   - Llama 3 : `https://huggingface.co/chat/`
   - For image : `https://stablediffusionweb.com/fr/app/image-generator`
3. Play with local LLM : ollama... But $>$ 4Gb per model
4. Benchmark (really difficult task $\Rightarrow$ human !)
   - `https://chat.lmsys.org`

Parametrization

- Choice of the LLM
  - Which size for which application?
  - Which budget?
  - Which ethical constraint?
  - Big + general + expensive *vs* small + adaptation + cheap
- Temperature

Parametrization

- Choice of the LLM
- Temperature



- Next word = sampling in a distribution
- Computation:
  word scores (logit) $ws\mathbb{R} \Rightarrow$ distrib. $p(w)$

$$p(w) = \frac{\exp(-\alpha T ws)}{\sum_i \exp(-\alpha T ws_i)}$$

- $T > 1 \Rightarrow$ Uniform = diversity
- $T < 1 \Rightarrow$ Dirac = Frozen/more deterministic

Getting started with LLM

`https://textsynth.com/completion.html`

- Distinguish LLM with/wo instruction
- Understand temperature

Compare instruct model *vs* pre-trained on a task, play with the temperature

What would be the story of a frog going in space?

Prompt & Information Access

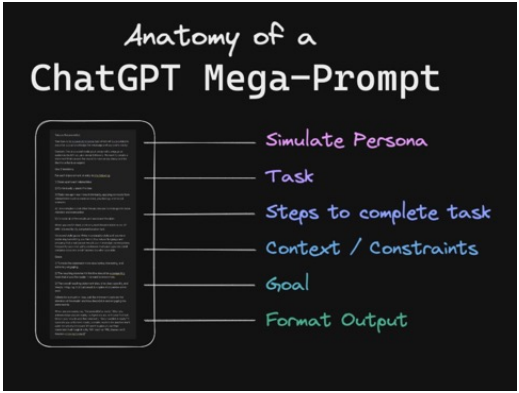
Relevant information = long prompt

Ranked by importance:

1. What is the question/task?
2. Who ask? / What is the context?
3. Steps to carry out the task
4. Output format / style
5. Exemple of pairs (questions/answers)

Solution 1: long prompt
Solution 2: Dialogue, ask several questions

Ordered when writing:



https://chatgptprompts.guru/what-makes-a-good-chatgpt-prompt/

Example (1) : Classical information access

- What is the definition of a *prompt*?

  In the field of LLMs, quickly explained with a simple vocabulary

- Try to write 5 lines about a domain in which you are not a specialist. Proposition:
  - What is *Conformal prediction*?
  - Differences between *Rationalism* and *Empiricism* in Enlightenment philosophy
  - Differences between a *metaphor* and an *allegory*?
  - What are the mainstays of the Mongolian economy?

- Let's ask the same question looking for a detailed answer for a specialist

Which risks? Which use? Which confidence?

Example (2) : NLP task

Given the sentence (from CONLL03):

`The European Commission said on Thursday it disagreed with German advice to consumers to shun British lamb until scientists determine whether mad cow disease can be transmitted to sheep.`

1. Extract the following entities with their types :*(place, person, organisation, date)*
2. Format the output in JSON
   - Is the result the same with formattting constraints?
3. Try some prompt from the appendix of GPTNER: `https://arxiv.org/pdf/2305.15444`
4. API Usage: `https://cookbook.openai.com/`

```
{
"entities": [
{
"type": "Organization",
"value": "European Commission"
},
...
```

Creativity, formatting & rephrasing

Brainstorming

- Build a detailed semester schedule for an advanced course in your field:
  - $\approx$ 10 courses
  - 2-3 parts in each courses
  - 5 to 10 bullet points in every part

- Obviously, you could do it: takes time + risk of forgetting things

$\Rightarrow$ Try to do it with LLMs

Motivation/recommendation letter

- Write a motivation letter for an MSc. in your field... And imagine your are undergraduate
  - Which keywords could you give at first? Which one could you discover?
  - Check your estimated strength wrt the targeted MSc.
  - Evaluate the generated letter as an examiner
- Write a recommendation letter for a student applying for a PhD in your field
  - How long do you know him? The course in which you met him? Main qualities...
  - + subject of the PhD

Meeting minutes

- Take some (non confidential) notes from a recent meeting

or write quickly some notes

or use this file:

1. Make the LLM write the meeting report

Summary

1. Generate a summary about the wikipedia page: `https://en.wikipedia.org/wiki/Variational_Bayesian_methods`
2. 10 lines, non technical vocabulary
3. 50 lines, more technical

4. Same thing with a recent news (for instance) `https://edition.cnn.com/2024/07/09/politics/biden-nato-analysis/index.html`

Text (article?) improvement

1. Get an article (your last one, your advisor's latest one...)
2. Copy-paste on paragraph
3. Add some noise (spelling mistake, bad formulation...)
4. Try to improve (optionnally translate)

⇒ What do you think about the result?

- Cheating?
- Simply exploiting new tools?
- Social rebalancing?

Conclusion

- LLM are not providing specific information
  - You have to give them to the system
  - Check what comes out first: these commonplace things are the same for everyone.
    - Keep the overall layout,
    - Personalize and reinforce qualities
- Ask the LLM for the list of qualities (& weaknesses) for a specific position

Is it cheating to use LLM for application/founding projects?

LLM are hard to detect... But it is easy to be caught!

See Guillaume Cabanac work on *Problematic Paper Screener*:
`https://www.irit.fr/~Guillaume.Cabanac/`

- Prompt forgetting :

#1 Statcheck comment accepted September 2016

Using the R package statcheck (v1.0.1), the HTML version of this article was scanned on 2016-08-05 for statistical results (t, r, F, Chi2, and Z values) reported in APA format (for specifics, see Nuijten et al., 2015). An automatically generated report follows.

The scan detected 9 statistical results in APA format, of which 0 contained potentially incorrect statistical results, of which 0 may change statistical significance (alpha = .05). Potential one-tailed results were taken into account when 'one-sided', 'one-tailed', or 'directional' occurred in the text.

Note that these are not definitive results and require manual inspection to definitively assess whether results are erroneous.

Reference
Nuijten, M. B., Hartgerink, C. H. J., van Assen, M. A. L. M., Epskamp, S., & Wicherts, J. M. (2015). The prevalence of statistical reporting errors in psychology (1985-2013). Behavior Research Methods.
http://dx.doi.org/10.3758/s13428-015-0664-2

`pubpeer.com`

- Tortured phrases
- Specific phrases : *...in the era of deep learning...*
- Math (!)

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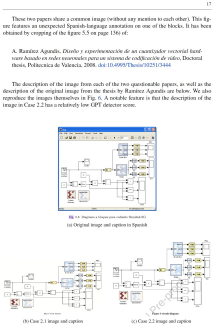

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Those two papers share a common image (without any mention to each other). This figure features an unexpected Spanish-language annotation on one of the blocks. It has been obtained by cropping of the figure 5.5 on page 136) of:

A. Ramírez Agundis, *Diseño y experimentación de un cuantizador vectorial basado en redes neuronales para un sistema de codificación de vídeo*, Doctoral thesis, Politécnica de Valencia, 2008. doi:10.4995/Thesis/10251/2444.

The description of the image from each of the two questionable papers, as well as the description of the original image from the thesis by Ramírez Agundis are below. We also reproduce the images themselves in Fig. 6. A notable feature is that the description of the image in Case 2.2 has a relatively low GPT detector score.

Code Generation

Getting started with a new library

- Start learning with simple examples
- Ask for Documentation

- Regarding machine learning and scikit learn library. Can you generate a python code that generate a 2 class toy example and compare linear and random forest classifer?
  - Ask the LLM to explain some part of the code !

- For instance: write a python/numpy short program that generate random point and plot them using `bokeh` so that index are displayed when the mouse passes over the point

- Generate a pink panther in html and visualize in your browser

Limitations

Bibliography

- Generate one bibliography on your favorite research topic
- Why the results won't be good?
- How may LLMs be used for this exercise?

Hard cases

- Animals & syllabus
  - Generate 10 animals the name of which count 4 syllabus
- Computations (divise, multiplies)
  - Compare direct computation (just ask for the result)
  - & code generation (ask for a python/R script to solve the problem)
- Reasonning (Lila dataset) `https://github.com/allenai/Lila`
  - Compare direct computation
  - & code generation

Run LLM Locally

OLlama: easy way to run locally

- LLM are huge and costly (both in computation & memory)
- ... But they have been dramatically optimized !
  - Quantization, pruning...

⇒ They can run locally on your machine

Simple solution: `ollama`: `https://ollama.com`

OLlama: easy way to run locally

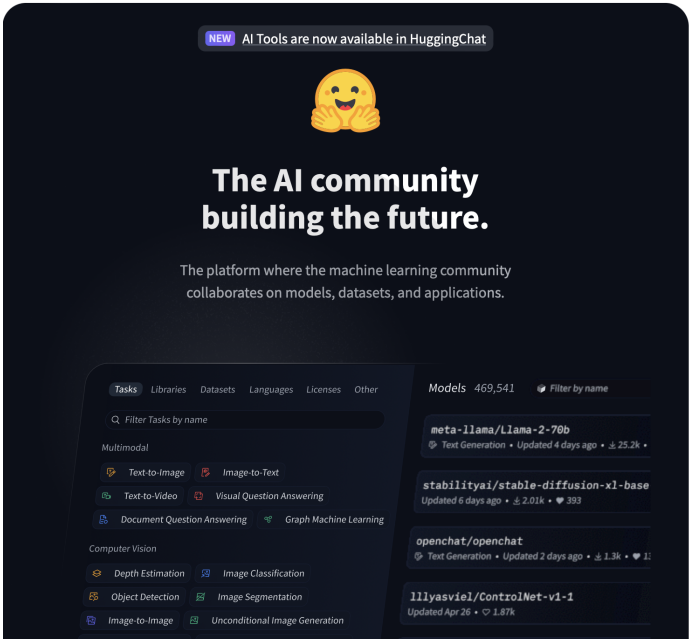
Here are some example models that can be downloaded:

Model	Parameters	Size	Download
Llama 3	8B	4.7GB	`ollama run llama3`
Llama 3	70B	40GB	`ollama run llama3:70b`
Phi 3 Mini	3.8B	2.3GB	`ollama run phi3`
Phi 3 Medium	14B	7.9GB	`ollama run phi3:medium`
Gemma 2	9B	5.5GB	`ollama run gemma2`
Gemma 2	27B	16GB	`ollama run gemma2:27b`
Mistral	7B	4.1GB	`ollama run mistral`
Moondream 2	1.4B	829MB	`ollama run moondream`
Neural Chat	7B	4.1GB	`ollama run neural-chat`
Starling	7B	4.1GB	`ollama run starling-lm`
Code Llama	7B	3.8GB	`ollama run codellama`
Llama 2 Uncensored	7B	3.8GB	`ollama run llama2-uncensored`
LLaVA	7B	4.5GB	`ollama run llava`
Solar	10.7B	6.1GB	`ollama run solar`

> Note: You should have at least 8 GB of RAM available to run the 7B models, 16 GB to run the 13B models, and 32 GB to run the 33B models.

Developments around LLM

HuggingFace



+Great tutorials
https://huggingface.co/learn/nlp-course/fr/chapter1/1

Example

```python
from transformers import pipeline
ner = pipeline("ner", grouped_entities=True)
ner("My name is Sylvain and I work at Hugging Face in Brooklyn.
```

```
[{'entity_group': 'PER',
'score': 0.9981694,
'word': 'Sylvain',
'start': 11,
'end': 18},
{'entity_group': 'ORG',
'score': 0.9796019,
'word': 'Hugging Face',
'start': 33,
'end': 45},
{'entity_group': 'LOC',
'score': 0.9932106,
'word': 'Brooklyn',
'start': 49,
'end': 57}]
```

Example (2)

```python
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
classifier([
"I've been waiting for a HuggingFace course my whole life.",
"I hate this so much!"
])
```

```
[{'label': 'POSITIVE', 'score': 0.9598049521446228},
{'label': 'POSITIVE', 'score': 0.996126115322113}]
```

Example (3)

```python
from transformers import pipeline

classifier = pipeline("text-generation")
result = classifier(["What is the color of the sun?"])
print(result)

result = classifier(["The sun color is "])
print(result)
```

...